Code Summary: CSS

The styles of the eight web tools in the Pi Zero 2 W build: how each stylesheet is stored, the design tokens it uses, how the light and dark themes work, and where the layout changes for smaller screens.

8 STYLESHEETS | DARK DEFAULT | STAGE 2

SUMMARY · READ-ONLY REVIEW

Where the styles are

The same eight pages as Stage 1. Their styles are not kept in one file. Each tool stores its CSS in two ways: a literal `<style>` block in the page, and Python string constants (names beginning `_CSS`) that are joined together when the page is served.

1.1 How the counts were made

- Both forms were collected for each tool: every `<style>` block in the file, and every string constant whose name contains CSS or THEME.
- Rule counts are approximate: they count selector blocks. Token counts are distinct custom properties (names beginning `--`).
- Nothing was rendered. Behaviour described here comes from the CSS text.

1.2 Two kinds of stylesheet

KIND	PAGES	SHAPE
One large sheet plus a small theme block	Node Controller, SysMon, wifimon, Pibrary	A single main sheet of 46,000 to 93,000 characters.
Many named parts joined at run time	instmon, 44helper, Node Talker, Node Recorder	Parts such as `_CSS_BASE`, `_CSS_LAYOUT` and `_CSS_COMPONENTS`, joined when the page is served.

Size and features per page

PAGE	PARTS	CHARACTERS	RULES	TOKENS	MEDIA RULES	ANIMATIONS	!IMPORTANT
Node Controller	2	85,901	468	56	11	3	2
SysMon	2	93,080	581	42	13	2	4
instmon	13	60,465	244	51	4	1	3
wifimon	2	46,394	159	40	2	0	4
44helper	12	59,889	256	43	2	0	2
Node Talker	5	60,475	346	32	4	1	95
Node Recorder	4	61,134	350	32	4	1	98
Pibrary	2	52,782	241	32	5	0	2

2.1 Breakpoints

Each page changes layout at a few widths. The list below gives the widths named in media queries, in pixels. Most pages have several small-screen steps.

PAGE	WIDTHS NAMED IN MEDIA QUERIES
Node Controller	560, 600, 900
SysMon	480, 600, 640, 800, 900, 1200
instmon	560, 600 (max-width) and 561 (min-width)
wifimon	964 (min-width)
44helper	560, 640
Node Talker	420, 480, 560
Node Recorder	420, 480, 560
Pibrary	600

wifimon's one layout rule is a min-width rule at 964 px, so it changes the layout for wider screens rather than narrower ones. The other pages adjust for narrower screens.

Light and dark

Seven of the eight pages share one theme mechanism. Pibrary has its own.

3.1 The shared mechanism

- The page is dark by default. A light theme is turned on by setting `data-theme="light"` on the root element.
- The choice is saved in the browser under `asl_dvs_theme`. If storage is not available, the page falls back to dark.
- The light theme is not a second set of colours. It applies `filter: invert(1) hue-rotate(180deg)` to the whole page, re-tunes the colour tokens for light, and then inverts images, video, canvas, iframes and checkboxes back to their original colours.
- Text shadows are removed in the light theme.
- A Dark and Light switch (a label and two buttons) appears in a settings area of each page. On 44helper it is on the Config tab. The active button is marked with `aria-pressed`.

3.2 What the approach means

- **One set of rules, two looks.** A rule written for dark applies to light without change. This keeps the CSS small.
- **Colours in images and icons.** Anything inverted must be inverted back, which the shared block does for the listed element types. A new image or canvas type would need the same treatment.
- **Cost on the Pi.** A filter on the whole page is applied to every element. Its cost on a Pi Zero 2 W has not been measured for this summary, and the book should not assume it is free.

3.3 Pibrary

Pibrary defines its own light and dark token sets, with no inversion. Its setting has three values: **dark**, **light** and **auto**. Auto follows the operating system through `prefers-color-scheme`. The value is saved under `pibrary.theme`. Pibrary is the only page that follows the operating system.

Design tokens

4.1 Where the tokens are

Each page defines its colours and sizes as custom properties on the root element, and the rules use them through `var(--name)`. The Node Controller defines 56 distinct tokens, and most of its colour tokens come in groups: a main colour, its RGB triple (for use in transparent fills) and a dimmed version. The groups are `--mc`, `--tc`, `--rc`, and others for mode colours.

4.2 Shared token names

Several pages use the same names for the same roles, so one theme block can serve them all. The names that recur include the accent colours (`--accent`, `--teal`, `--red`, `--green`, `--amber`, `--blue`, `--orange`), and the text and surface colours. The light theme block re-tunes these names, not the rules that use them.

4.3 Pibrary's tokens

Pibrary uses its own names. Its dark set includes `--bg`, `--surface`, `--surface2` and `--border`, and its light set uses the same names with light values.

Node Controller and SysMon

5.1 Node Controller

- Two parts: a small block for the theme (1,327 characters) and the main sheet (84,573 characters).
- 56 tokens, and 468 rules, the most of any page after SysMon.
- Eleven media rules, including one for `prefers-reduced-motion`, so its animations can be turned off for users who ask for less motion.
- `z-index` values run from 100 to 9,999, and the highest value is 9,999.
- Three animations.

5.2 SysMon

- The largest sheet of the eight: 93,080 characters and 581 rules.
- Thirteen media rules, at 480, 600, 640, 800, 900 and 1200 px, to adjust the layout as the screen narrows.
- Eight layers, from 200 to 9,999.
- Four `!important` rules, and two animations. No reduced-motion rule.

wifimon, 44helper and instmon

6.1 wifimon

- One main sheet of 45,066 characters and 159 rules: the smallest rule count of the large sheets, which suits a page of cards.
- The one layout rule is at 964 px (min-width), so the card grid widens on larger screens rather than narrowing on smaller ones.
- Respects reduced motion, with one rule.

6.2 44helper

- Twelve parts: a base sheet of 33,082 characters, a layout part, a components part, and one part for each section (nodes, firewall, logs, ASL3, auth and quiet mode).
- Media rules at 560 and 640 px only.
- No reduced-motion rule and no animation keyframes.

6.3 instmon

- Thirteen parts, one for each section of the main page (base, layout, components, library, log, editor, login and disk image), plus the theme block.
- Media rules at 560 and 600 px (max-width), one at 561 px (min-width), and one reduced-motion rule.
- The Suite Log page (Stage 1) uses its own short sheet inside its page, not the shared parts.

Node Talker, Node Recorder and Pibrary

7.1 Node Talker

- A large part of 53,259 characters, with four smaller parts in the page.
- It uses `!important` 95 times. Many of these force a layout at phone widths: for example, the connect bar is made full width and its buttons share a row.
- Media rules at 420, 480 and 560 px. Its small-screen focus fits a phone-first page.
- No reduced-motion rule. Its localStorage use is the heaviest of the eight (29 references in the page code), which is a JavaScript matter for Stage 3.

7.2 Node Recorder

- A large part of 53,968 characters.
- It is a near copy of Node Talker's sheet. The two differ in about 48 lines (for example, the Node Recorder has a JSON panel and a conversion terminal, and the listen-row rules are absent). Changes to one do not reach the other.
- `!important` is used 98 times, in the same pattern.

7.3 Pibrary

- One main sheet of 52,773 characters and 241 rules, with few `!important` uses (2).
- Five media rules, all at 600 px.
- Its own theme (Part 3.3). It is the only page with a dark, light and auto choice.

Things the stylesheets show

8.1 Copies drift apart

Node Talker and Node Recorder share most of their stylesheet, but have separately edited copies. A fix in one will not reach the other. The book should describe the two as related, not shared.

8.2 Heavy use of `!important`

Node Talker (95) and Node Recorder (98) use `!important` far more than any other page. These rules are hard to override, so a new style for those pages may have to fight them. Document the reason before a change is made.

8.3 Reduced motion is not universal

Only the Dashboard, wifimon and instmon respond to a request for less motion. The other pages animate regardless. Check before the book claims the pages respect that setting.

8.4 The light theme is a filter

The light theme of seven pages depends on a whole-page filter. It is simple, but it needs care with images and new elements, and its cost on the Pi Zero 2 W has not been measured. The book should say so.

8.5 Inline CSS, no shared file

No page loads a stylesheet file. Every rule is served inside the page. A change to a page's look means editing its tool file, and the change ships with that tool's version.